

Representation Theory

Labix

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Abstract

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1 Representation of Groups

1.1 Basic Definitions

Definition 1.1.1 (Representations)

Let G be a group. Let k be a field. A representation of G is a k -vector space V together with a group homomorphism

$$\rho : G \rightarrow \text{End}_k(V)$$

Example 1.1.2

The following are representations of $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$.

- $\rho : D_8 \rightarrow GL_2(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$.
- $\rho : D_8 \rightarrow GL_n(\mathbb{C})$ defined by $\rho(r) = \rho(s) = I_n$.
- $\rho : D_8 \rightarrow \mathbb{C}^\times$ defined by $\rho(r) = \pm 1$ and $\rho(s) = \pm 1$.
- $\rho : D_8 \rightarrow GL_4(\mathbb{C})$ defined by $\rho(r) = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $\rho(s) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

Lemma 1.1.3

Let G be a group. Let k be a field. Then there is a one to one correspondence

$$\left\{ \rho : G \rightarrow \text{End}_k(V) \mid \begin{array}{c} \rho \text{ is a} \\ \text{representation of } G \end{array} \right\} \xleftrightarrow{1:1} \left\{ \rho : k[G] \rightarrow \text{End}_k(V) \mid \begin{array}{c} \rho \text{ is a } k\text{-algebra} \\ \text{homomorphism} \end{array} \right\}$$

given by the universal property of the group ring $k[G]$.

Lemma 1.1.4

Let G be a group. Let k be a field. Then there is a one to one correspondence

$$\left\{ \rho : k[G] \rightarrow \text{End}_k(V) \mid \begin{array}{c} \rho \text{ is a } k\text{-algebra} \\ \text{homomorphism} \end{array} \right\} \xleftrightarrow{1:1} \left\{ f : k[G] \times V \rightarrow V \mid \begin{array}{c} f \text{ defines an } k[G]\text{-module} \\ \text{structure on } V \end{array} \right\}$$

given by $\rho \mapsto ((g, v) \mapsto \rho(g)(v))$.

Definition 1.1.5 (Types of Representations)

Let G be a group. Let k be a field. Let V be a representation of G .

- We say that V is irreducible if it is simple as a $k[G]$ -module.
- We say that V is semisimple if it is semisimple as a $k[G]$ -module.
- We say that V is indecomposable if it is indecomposable as a $k[G]$ -module.

Example 1.1.6

The representation of $C_3 = \langle x \mid x^3 \rangle$ on $\mathbb{F}_3^2$ given by

$$\rho(x) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

is not semisimple.

Definition 1.1.7 (Composition Factors)

Let G be a group. Let k be a field. Let V be a semisimple representation of G . Define the composition factors of V to be

$$\text{Comp}(V) = \{V_1, \dots, V_n\}$$

where the irreducible decomposition of V is given by

$$V \cong \bigoplus_{i=1}^n V_i$$

Definition 1.1.8 (Multiplicity of an Irreducible Submodule)

Let G be a group. Let k be a field. Let V be a semisimple representation of G . Let W be an irreducible representation of V . Define the multiplicity of W in V to be

$$\text{mult}_W(V) = |\{V_i \in \text{Comp}(V) \mid V_i \cong W \text{ as } k[G]\text{-modules}\}|$$

Definition 1.1.9 (The Set of (Irreducible) Representations)

Let G be a group. Let k be a field.

- Define the set of representations of G by

$$\text{Rep}(G) = \{[V] \mid V \text{ is a representation of } G\}$$

- Define the set of representations of G by

$$\text{IrrRep}(G) = \{[V] \mid V \text{ is an irreducible representation of } G\}$$

Definition 1.1.10 (The Regular Representations)

Let G be a group. Let k be a field. Define the regular representation of G to be the function $\text{reg}_G : G \rightarrow \text{End}_k(k[G])$ given by

$$\rho(g)(h) = g \cdot h$$

Thus the regular representation of G is just the $k[G]$ -algebra $k[G]$.

Definition 1.1.11 (The Permutation Representations)

Let X be a set. Let G be a group acting on X . Let k be a field. Define the permutation representation $\rho : G \rightarrow \text{GL}(k\langle X \rangle)$ of G on X over k by the formula

$$\rho(g)(x) = g \cdot x$$

and then extending it k -linearly.

The regular representation is just the permutation representation with the group action on itself.

Definition 1.1.12 (G -Invariant Subspaces)

Let G be a group. Let k be a field. Let V be a representation of G . Define the G -invariant subspace of V to be

$$V^G = \{v \in V \mid \rho(g)(v) = v \text{ for all } g \in G\}$$

If we think of representations as actions of $k[G]$ on V and hence actions of G on V , then V^G is just the fixed points of the action.

1.2 Maps between Representations

Definition 1.2.1 (Homomorphisms of Representations)

Let G be a group. Let k be a field. Let $\rho : G \rightarrow \text{End}_k(V)$ and $\theta : G \rightarrow \text{End}_k(W)$ be representations of G . Let $f : V \rightarrow W$ be a function. We say that f is a homomorphism of representations if the following are true.

- f is a k -linear map.
- $f \circ \rho(g) = \theta(g) \circ f$ for all $g \in G$.

Lemma 1.2.2

Let G be a group. Let k be a field. Let V, W be representations of G . Let $f : V \rightarrow W$ be a function. Then f is a homomorphism of representations if and only if f is a $k[G]$ -algebra homomorphism.

Definition 1.2.3 (Isomorphisms of Representations)

Let G be a group. Let k be a field. Let V, W be representations of G . Let $f : V \rightarrow W$ be a homomorphism. We say that f is an isomorphism if there exists a homomorphism $g : W \rightarrow V$ such that

$$g \circ f = \text{id}_V \quad \text{and} \quad f \circ g = \text{id}_W$$

In this case we say that V and W are isomorphic representations.

Lemma 1.2.4

Let G be a group. Let k be a field. Let V, W be representations of G . Let $f : V \rightarrow W$ be a homomorphism. Then f is an isomorphism if and only if f is an isomorphism of k -vector spaces.

Let R be a ring. Let M, N be R -modules. Recall that in general there is no canonical module structure for $\text{Hom}_R(M, N)$ unless R is commutative. Hence $\text{Hom}_{k[G]}(V, W)$ is a priori not a representation of G . However, we will give a module structure to it by giving a $k[G]$ -module structure to $\text{Hom}_k(V, W)$ and identifying $\text{Hom}_{k[G]}(V, W)$ with $\text{Hom}_k(V, W)^G$.

Lemma 1.2.5

Let G be a group. Let k be a field. Let $\rho_V : G \rightarrow \text{End}_k(V)$ and $\rho_W : G \rightarrow \text{End}_k(W)$ be representations of G . Then $\text{Hom}_k(V, W)$ is also a representation of G via the map $g \mapsto \rho_W(g) \circ - \circ \rho_V(g)^{-1}$.

Proposition 1.2.6

Let G be a group. Let k be a field. Let V, W be representations of G . Then there is a vector space isomorphism

$$\text{Hom}_{k[G]}(V, W) \cong \text{Hom}_k(V, W)^G$$

given by restriction of scalars.

Proof

Let $f : V \rightarrow W$ be a $k[G]$ -module homomorphism. We claim that after restricting to k , we have $f \in \text{Hom}_k(V, W)^G$. Indeed, since f is a homomorphism of representations, we have $\rho_W(g) \circ f = f \circ \rho_V(g)$ for all $g \in G$. This implies that $\rho_{\text{Hom}_k(V, W)}(g)(f) = \rho_W(g) \circ f \circ \rho_V(g)^{-1} = f$. Hence $f \in \text{Hom}_k(V, W)^G$. Hence we have a well defined k -linear map $\text{Hom}_{k[G]}(V, W) \rightarrow \text{Hom}_k(V, W)^G$. This map is clearly injective and so is a vector space isomorphism. ■

Note that the above isomorphism of vector spaces give $\text{Hom}_{k[G]}(V, W)$ the structure of a $k[G]$ -module.

1.3 One Dimensional Representations

Let G be a group. A one dimensional representation of G is a group homomorphism $G \rightarrow k^\times$. Since $k^\times$ is abelian, we may invoke the universal property of the abelianization G^{ab} in Advanced Group Theory to see that there is a one to one bijection between one dimensional representations of G and one dimensional representations G^{ab} .

Lemma 1.3.1

Let G be a group. If G^{ab} is finite, then the number of one dimensional representations of G is G^{ab} .

Let G be a group. Recall that G is perfect if $G = [G, G]$. Equivalently, this is the same as saying that $G^{\text{ab}} = 0$.

Lemma 1.3.2

Let G be a group. If G is perfect, then there are no non-trivial one dimensional representations of G .

Thus simple groups have no non-trivial one dimensional representations.

Proposition 1.3.3

Let G be a group. Let k be a field. Let $\rho : G \rightarrow \text{End}_k(V)$ be a representation of G . Let $\theta : G \rightarrow k^\times$ be a one dimensional representation of G . Then the following are true.

- $\theta\rho : G \rightarrow \text{End}_k(V)$ is also a representation of G .
- $\theta\rho$ is irreducible if and only if ρ is irreducible.
- $\theta\rho$ is the tensor product of θ and ρ as $k[G]$ -modules.

Proof

Let $g, h \in G$. Then we have

$$(\theta\rho)(gh) = \theta(gh)\rho(gh) = \theta(g)\theta(h)\rho(g)\rho(h) = \theta(g)\rho(g)\theta(h)\rho(h) = (\theta\rho)(g)(\theta\rho)(h)$$

since $\theta(h) \in k^\times$ is a scalar. Hence $\theta\rho$ is a representation.

Suppose that ρ has a subrepresentation W . Then for all $w \in W$, we have $(\theta\rho)(g)(w) = \theta(g)(w)\rho(g)(w) \in W$ since $\theta(g)(w) \in k^\times$ is a scalar and W is a vector space. Hence W is a subrepresentation of $\theta\rho$. Conversely, notice that if $\theta : G \rightarrow k^\times$ is a representation, then $(-)^{-1} \circ \theta : G \rightarrow k^\times$ is also a one dimensional representation. Thus if W is a subrepresentation of $\theta\rho$, then W is a subrepresentation of $((-)^{-1} \circ \theta)\theta\rho = \rho$.

The third item follows from the fact that $k \otimes_k V \cong V$ as vector spaces. ■

1.4 Representations over Algebraically Closed Fields

Lemma 1.4.1 (Schur's Lemma III)

Let G be a group. Let k be an algebraically closed field. Let V, W be irreducible representations of G . Let $\phi : V \rightarrow W$ be a homomorphism. Then the following are true.

- If $V \cong W$ and they are finite dimensional, then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.
- If $V \not\cong W$, then $\phi = 0$.

Corollary 1.4.2

Let G be a group. Let k be an algebraically closed field. Let V be a finite dimensional representation of G . If G is abelian, then $\dim_k(V) = 1$.

Corollary 1.4.3

Let G be a group. Let k be an algebraically closed field. Let V, W be finite dimensional irreducible representations of G . Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Corollary 1.4.4

Let G be a group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G such that W is irreducible. Then we have

$$\dim_k(\text{Hom}_{k[G]}(V, W)) = \text{mult}_W(V)$$

1.5 Representations of Finite Groups over Algebraically Closed Fields

Landmark result: Maschke's theorem. Recall that it says that $k[G]$ is a semisimple k -algebra if G is finite and $\text{char}(k)$ does not divide $|G|$. By a result in Rings and Modules, this implies that every $k[G]$ -module is semisimple. Hence every representation of G is semisimple.

Proposition 1.5.1

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Suppose that

$$k[G] \cong \bigoplus_{i=1}^n \text{End}_k(V_i)$$

be the decomposition given by the Artin-Wedderburn theorem. Then we have

$$\text{IrrRep}(G) = \{[V_1], \dots, [V_n]\}$$

Proof

Follows from a corollary of Artin-Wedderburn theorem in Advanced Ring Theory. ■

Corollary 1.5.2

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Then we have

$$|G| = \sum_{V \in \text{IrrRep}(G)} (\dim_k(V))^2$$

Proof

Follows from the above proposition. ■

Example 1.5.3

Let G be a group of order 8. Let k be an algebraically closed field such that $\text{char}(k) \neq 2$. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible $k[G]$ -modules. Then exactly one of the following is true (up to reordering of the modules).

- $n = 8$ and $\dim(V_i) = 1$.
- $n = 5$ and $\dim(V_1) = \dots = \dim(V_4) = 1$ and $\dim(V_5) = 2$.

Corollary 1.5.4

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. If $[V] \in \text{IrrRep}(G)$ is an irreducible representation, then we have

$$\dim_k(V) < \sqrt{|G|}$$

Proof

We have

$$|G| = \sum_{V \in \text{IrrRep}(G)} (\dim_k(V))^2 > \dim_k(V)^2$$

where the inequality is strict because the trivial representation is irreducible and always exists. ■

2 Character Theory

2.1 Basic Definitions

Definition 2.1.1 (Characters)

Let G be a finite group. Let k be an algebraically closed field. Let $\rho : G \rightarrow \text{End}_k(V)$ be a finite dimensional representation of G . Define the character of V to be the function $\chi_V : G \rightarrow k$ given by

$$\chi_V(g) = \text{tr}(\rho(g))$$

Proposition 2.1.2

Let G be a finite group. Let k be an algebraically closed field such that $\text{char}(k)$ does not divide $|G|$. Let $\rho : G \rightarrow \text{End}_k(V)$ be a representation of G . Then the following are true.

- $\chi_V(1_G) = \dim(V)$.
- $\chi_V(hgh^{-1}) = \chi_V(g)$ for all $g, h \in G$.

Proof

By Linear Algebra, we have that

$$\chi_V(1_G) = \text{tr}(\rho(1_G)) = \text{tr}(\text{id}_V) = \text{tr}(I_{\dim(V)}) = \dim(V)$$

so $\chi_V(1_G) = \dim(V)$.

Let $g, h \in G$. Then we have

$$\chi_V(hgh^{-1}) = \text{tr}(\rho(h)\rho(g)\rho(h^{-1})) = \text{tr}(\rho(h)\rho(h^{-1})\rho(g)) = \text{tr}(\rho(hh^{-1})\rho(g)) = \text{tr}(\rho(g)) = \chi_V(g)$$

by properties of the trace map. ■

The second item implies that χ descends to a function on conjugation classes $\chi : \text{Cl}(G) \rightarrow k$.

Lemma 2.1.3

Let G be a finite group. Let k be an algebraically closed field. Then χ descends to a well defined class function $\chi : \text{Cl}(G) \rightarrow k$.

Lemma 2.1.4

Let G be a finite group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G . Then the following are true.

- If $W \leq V$ is a subrepresentation, then $\chi_V = \chi_W + \chi_{V/W}$.
- $\chi_{V \otimes W} = \chi_V \chi_W$.
- $\chi_{V^*} = \overline{\chi}$.
- $\chi_{\text{Hom}_k(V, W)} = \overline{\chi_V} \chi_W$.

Lemma 2.1.5

Let G be a finite group. Let k be an algebraically closed field. Let V, W be finite dimensional representations of G . If $V \cong W$ are isomorphic, then we have

$$\chi_V = \chi_W$$

Proposition 2.1.6

Let G be a finite group. Let $\rho : G \rightarrow \text{End}_{\mathbb{C}}(V)$ be a complex representation of G . Let $g \in G$. Then the following are true.

- Let ω be a $|g|$ th root of unit. Then we have

$$\chi(g) = \sum_{i=1}^k \omega^{j_i}$$

- for some $j_i \in \mathbb{N}$ and $k \in \mathbb{N}$.
- We have $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

Proof

Let $H = \langle g \rangle$. Then V is also a $\mathbb{C}[H]$ -module. Let $V = \bigoplus_{i=1}^m V_i$ be the decomposition of V into irreducible $\mathbb{C}[H]$ -submodules. Since H is cyclic, we must have $\dim(V_i) = 1$ by crl2.1.2. Choose non-zero $v_i \in V_i$. Then $E = \{v_1, \dots, v_m\}$ is a basis for V and $[\rho(g)]_E^E$ is a diagonal matrix. Now for each i , V_i is a $\mathbb{C}[H]$ -module. Hence $\rho(g)(v_i) \in V_i$ implies that $\rho(g)(v_i) = \lambda_i v_i$ for some $\lambda_i \in \mathbb{C}^\times$. Then we have $\lambda^{|g|} v_i = \rho(g)^{|g|}(v_i) = v_i$ so that λ is a $|g|$ th root of unity. Since $\rho(g)$ is diagonal, we conclude that the trace of $\rho(g)$ is a sum of $|g|$ th roots of unity.

From the above proof, we know that $[\rho(g)]_E^E = \text{diag}(\omega^{-j_1}, \dots, \omega^{-j_m}) = \text{diag}(\overline{\omega^{j_1}}, \dots, \overline{\omega^{j_m}})$. Hence $\chi_V(g^{-1}) = \overline{\chi_V(g)}$. ■

Definition 2.1.7 (Irreducible Characters)

Let G be a finite group. Let k be an algebraically closed field. Let V be a finite dimensional representation of G . We say that χ_V is irreducible if V is irreducible.

Definition 2.1.8 (Characters of a Group)

Let G be a finite group. Let k be an algebraically closed field. Let $\chi : G \rightarrow k$ be a function. We say that χ is a character of G if there exists a finite dimensional representation V of G such that $\chi = \chi_V$.

2.2 Orthogonality of Characters of Irreducible Representations

Definition 2.2.1 (The Inner Product Space of Class Functions)

Let G be a finite group. Define the inner product space of class functions of G to be the $\mathbb{C}$ -vector space

$$\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$$

together with the inner product given by

$$\langle f_1, f_2 \rangle = \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

for $f_1, f_2 \in \text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$.

Proposition 2.2.2

Let G be a finite group. Let V, W be finite representations of G over $\mathbb{C}$. Then we have

$$\langle \chi_V, \chi_W \rangle = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V, W))$$

Proof

Step 1:

Let $T : V \rightarrow V^G$ be the map defined by $v \mapsto \frac{1}{|G|} \sum_{g \in G} \rho(g)(v)$. I claim that $\text{tr}(T) = \dim_{\mathbb{C}}(V^G)$. Notice that since V^G is G -invariant, we have that $T^2 = T$. I claim that $V^G = \ker(T) \oplus \text{im}(T)$. Suppose that $v \in \ker(T) \cap \text{im}(T)$. Then there exists $w \in V$ such that $T(w) = v$. Now $v = T(w) = T^2(w) = T(T(w)) = T(v) = 0$. Hence $\ker(T)$ and $\text{im}(T)$ has empty intersection. Now let $v \in V$. Then we have $v = (v - T(v)) + T(v)$. The latter lies in $\text{im}(T)$ and the former lies in $\ker(T)$ since $T(v - T(v)) = T(v) - T(T(v)) = T(v) - T(v) = 0$. Hence $V = \ker(T) \oplus \text{im}(T)$. Choose a basis $b_1, \dots, b_k$ for $\text{im}(T)$, choose a basis for $b_{k+1}, \dots, b_n$ for $\ker(T)$. Since $V = \ker(T) \oplus \text{im}(T)$, matrix for T under the basis $b_1, \dots, b_n$ is given by $\begin{pmatrix} I_{\dim(\text{im}(T))} & 0 \\ 0 & 0 \end{pmatrix}$. Hence $\text{tr}(T) = \dim(\text{im}(T)) = \dim(V^G)$.

Step 2:

We have

$$\langle \chi_{\text{triv}}, \chi_V \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_V(g)} = \frac{1}{|G|} \sum_{g \in G} \overline{\text{tr}(\rho(g))} = \text{tr} \left(\frac{1}{|G|} \sum_{g \in G} \overline{\rho(g)} \right) = \text{tr}(\overline{T}) = \overline{\text{tr}(T)} = \dim_{\mathbb{C}}(V^G)$$

Step 3:

We have

$$\langle \chi_V, \chi_W \rangle = \frac{1}{|G|} \sum_{g \in G} \chi_V(g) \overline{\chi_W(g)} = \frac{1}{|G|} \sum_{g \in G} 1 \cdot \overline{\chi_V(g) \chi_W(g)} = \langle 1, \overline{\chi_V} \chi_W \rangle = \langle \chi_{\text{triv}}, \chi_{\text{Hom}_{\mathbb{C}}(V, W)} \rangle$$

Step 4:

Combining all the above, we have

$$\langle \chi_V, \chi_W \rangle = \langle \chi_{\text{triv}}, \chi_{\text{Hom}_{\mathbb{C}}(V, W)} \rangle = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}}(V, W)^G) = \dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V, W))$$

■

Corollary 2.2.3

Let G be a finite group. Let V, W be finite representations of G over $\mathbb{C}$. Then the following are true.

- If V and W are irreducible, then we have

$$\langle \chi_V, \chi_W \rangle = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

- V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.

Proof

Follows directly from the properties of $\dim_{\mathbb{C}}(\text{Hom}_{\mathbb{C}[G]}(V, W))$.

■

Proposition 2.2.4

Let G be a finite group. Then the following are true.

- We have $|\text{IrrRep}(G)| = \dim_{\mathbb{C}}(\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})) = |\text{Cl}(G)|$.
- The set $\{\chi_V \mid V \in \text{IrrRep}(G)\}$ is an orthonormal basis for $\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$.

Proof

I claim that $\dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = |\text{Cl}(G)|$. Let $x = \sum_{h \in G} \lambda_h h \in \mathbb{C}[G]$. Notice that $xy = yx$ for all $y \in \mathbb{C}[G]$ if and only if $xg = gx$ for all $g \in G$ by linearity. Now notice that we have $gxg^{-1} = x$ if and only if

$$\sum_{h \in G} \lambda_h ghg^{-1} = \sum_{h \in G} \lambda_h h$$

By comparing basis for $\mathbb{C}[G]$, we see that this is true if and only if $\lambda_{g^{-1}hg} = \lambda_h$ for all $h \in G$. Thus $x = \sum_{h \in G} \lambda_h h \in Z(\mathbb{C}[G])$ if and only if $\lambda_{g^{-1}hg} = \lambda_h$ for all $h \in G$. This means that x is constant on conjugacy classes. Such a vector space has dimension $|\text{Cl}(G)|$.

Now by Artin-Wedderburn theorem, we have

$$\mathbb{C}[G] \cong \bigoplus_{V \in \text{IrrRep}(G)} \text{End}_{\mathbb{C}}(V)$$

Since taking center respects products, we have

$$Z(\mathbb{C}[G]) \cong \bigoplus_{V \in \text{IrrRep}(G)} Z(\text{End}_{\mathbb{C}}(V)) \cong \mathbb{C}^{|\text{IrrRep}(G)|}$$

since $\text{End}_{\mathbb{C}}(V) \cong M_{\dim(V)}(\mathbb{C})$ and from advanced ring theory we have seen that $Z(M_{\dim(V)}(\mathbb{C})) \cong \mathbb{C}$. Taking dimensions on both sides, we deduce that $|\text{Cl}(G)| = \dim_{\mathbb{C}}(Z(\mathbb{C}[G])) = |\text{IrrRep}(G)|$.

Now since $\{\chi_V \mid V \in \text{IrrRep}(G)\}$ are orthogonal, they are linearly independent and has the same cardinality the dimension of $\text{Hom}_{\text{Set}}(\text{Cl}(G), \mathbb{C})$. Hence they form an orthonormal basis. ■

Corollary 2.2.5

Let G be a finite group. Let $g, h \in G$. Then g and h are conjugate if and only if $\chi(g) = \chi(h)$ for all characters χ of G .

Corollary 2.2.6

Let G be a finite group. Let $g \in G$. Then g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ of G .

2.3 Character Tables

Definition 2.3.1 (The Character Table of a Finite Group)

Let G be a finite group. Let $V_1, \dots, V_{|\text{Cl}(G)|}$ be the complete list of non-isomorphic irreducible $\mathbb{C}[G]$ -modules. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G . Define the character table of G to be the matrix $(a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ defined by

$$a_{i,j} = \chi_{V_i}(g_j)$$

Proposition 2.3.2

Let G be a finite group. Let $\text{Cl}(g_1), \dots, \text{Cl}(g_{|\text{Cl}(G)|})$ be the complete list of distinct conjugacy classes of G . Let $A = (a_{i,j}) \in M_{|\text{Cl}(G)|}(\mathbb{C})$ be the character table of G . Then the following are true.

- The rows of A are orthogonal:

$$\frac{1}{|G|} \sum_{j=1}^{|\text{Cl}(G)|} |\text{Cl}(g_j)| \chi_{V_r}(g_j) \overline{\chi_{V_s}(g_j)} = \sum_{j=1}^{|\text{Cl}(G)|} |\text{Cl}(g_j)| a_{r,j} \overline{a_{s,j}} = \begin{cases} 1 & \text{if } r = s \\ 0 & \text{otherwise} \end{cases}$$

- The columns of A are orthogonal. This means that

$$\sum_{i=1}^{|\text{Cl}(G)|} \chi_{V_i}(g_r) \overline{\chi_{V_i}(g_s)} = \sum_{i=1}^{|\text{Cl}(G)|} a_{i,r} \overline{a_{i,s}} = \begin{cases} |\text{Cl}(g_r)| & \text{if } r = s \\ 0 & \text{otherwise} \end{cases}$$

A number of previous theorems help determining the character table. To compute the number of irreducible representations, we may use the following facts.

- $|G| = \sum_{V \in \text{IrrRep}(G)} \dim(V)^2$.
- $|\text{Cl}(G)| = |\text{IrrRep}(G)|$.

To then figure out the character table, we may use the following facts.

- $\chi_V(1_G) = \dim(V)$.
- The trivial map $G \rightarrow \mathbb{C}$ given by $g \mapsto 1$ is always a representation of G corresponding to the trivial representation.
- Orthogonality relations between the rows and the columns
- $\chi(g^{-1}) = \overline{\chi(g)}$ for all characters χ .
- g and g^{-1} are conjugate if and only if $\chi(g) \in \mathbb{R}$ for all characters χ .
- If χ is an irreducible character, then $\overline{\chi}$ is also an irreducible character corresponding to the dual representation.
- If V is one-dimensional, then $\chi_V = \rho$ and χ is a group homomorphism.

Example 2.3.3

The character table for $D_6 = \langle r, s \mid r^3, s^2, rsrs \rangle$ is given by

D_6	1	$\{r, r^2\}$	$\{s, sr, sr^2\}$
χ_{Trivial}	1	1	1
χ_2	1	1	-1
χ_3	2	-1	0

where χ_2 is the character associated to the representation $D_6 \rightarrow \mathbb{C}$ given by $r \mapsto 1, s \mapsto -1$ and χ_3 is the character associated to the representation

Example 2.3.4

The character table for $D_8 = \langle r, s \mid r^4, s^2, rsrs \rangle$ is given by

D_8	1	$\{r, r^3\}$	$\{r^2\}$	$\{s, sr^2\}$	$\{sr, sr^3\}$
χ_{Trivial}	1	1	1	1	1
χ_2	1	1	1	-1	-1
χ_3	1	-1	1	1	-1
χ_4	1	-1	1	-1	1
χ_5	2	0	-2	0	0

Example 2.3.5

The character table for A_4 is given by

A_4	1	(1 2) (3 4)	(1 2 3) (1 3 2)
χ_{Trivial}	1	1	1
χ_2	1	1	$e^{2\pi i/3}$
χ_3	1	1	$e^{4\pi i/3}$
χ_3	3	-1	0

where χ_2 is the character associated to the representation $D_6 \rightarrow \mathbb{C}$ given by $r \mapsto 1, s \mapsto -1$ and χ_3 is the character associated to the representation

Example 2.3.6

The character table for C_n is given by the matrix:

$$\begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & \zeta & \cdots & \zeta^{n-1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \zeta^{n-1} & \cdots & (\zeta^{n-1})^{n-1} \end{pmatrix}$$

where $\zeta = e^{2\pi i/n}$.

2.4 Representation Theory of Subgroups

Recall that we have the notion of restriction of scalars and extension of scalars in Rings and Modules. Let G be a group. Let k be a field. Let $H \leq G$ be a subgroup.

- Let V be a representation of G . The restriction of V to H is the k -vector space

$$\text{Res}_H^G(V) = V$$

together with the action of $k[H]$ given by restricting the action of $k[G]$ to $k[H]$.

- Let U be a representation of H . The extension of U to G is the k -vector space

$$\text{Ind}_H^G(U) = k[G] \otimes_{k[H]} U$$

together with the action of $k[G]$ given by the action on the component $k[G]$ in the tensor product.

Note: Even if V is an irreducible $k[G]$ -module, it may not be an irreducible $k[H]$ -module, and the reverse is also true.

Proposition 2.4.1

Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a non-zero character of H . Then there exists an irreducible character χ of G such that $\langle \text{Res}_H^G \chi, \psi \rangle \neq 0$.

Proof

Let $\chi_1, \dots, \chi_k$ be the complete list of irreducible characters of G . Let χ_{reg} be the character of the regular representation of G . Then we know that $\chi_{\text{reg}} = \sum_{i=1}^k \chi_i(1_G) \chi_i$. Notice that the regular $\mathbb{C}[G]$ -module contains $\mathbb{C}[H]$ and $\mathbb{C}[H]$ is closed under the action of H . Hence $\mathbb{C}[H]$ is a submodule of $\text{Res}_H^G(\mathbb{C}[G])$.

Suppose that ψ is irreducible. Then the corresponding $\mathbb{C}[H]$ -module V is a submodule of the regular representation $\mathbb{C}[H]$. Thus $\langle \text{Res}_H^G \chi_{\text{reg}}, \psi \rangle \neq 0$. Then we have

$$0 \neq \left\langle \sum_{i=1}^k \chi_i(1_G) \text{Res}_H^G \chi_i, \psi \right\rangle = \sum_{i=1}^k \chi_i(1_G) \langle \text{Res}_H^G \chi_i, \psi \rangle$$

Thus there exists χ_i such that $\langle \text{Res}_H^G \chi_i, \psi \rangle \neq 0$.

We can extend this to the case where ψ is reducible by noting that for a character ψ_j for some irreducible composition factor of ψ , we have $\langle \text{Res}_H^G \chi_i, \psi_j \rangle \neq 0$. ■

Theorem 2.4.2 (Clifford's Theorem)

Let G be a finite group. Let $H \trianglelefteq G$ be a normal subgroup. Let χ be an irreducible character of G . Let $\psi_1, \dots, \psi_n$ be the complete list of irreducible characters of H such that $\text{Res}_H^G \chi$ is a linear combination of $\psi_1, \dots, \psi_n$ with non-zero coefficients. Then the following are true.

- Each ψ_i have the same dimension.
- $\langle \text{Res}_H^G \chi, \psi_i \rangle$ is the same for all i .

Proof

Let V be an irreducible $\mathbb{C}[G]$ -module corresponding to χ . Let U be an irreducible $\mathbb{C}[H]$ -module of $\text{Res}_H^G(V)$. For all $g \in G$, the set $g \cdot U$ is a subspace of V . For all $h \in H$, we have

$$h(gu) = g(g^{-1}hg)u \in gU$$

since U is a $\mathbb{C}[H]$ -module and H is normal implies $g^{-1}hg \in H$. So $g \cdot U$ is a $\mathbb{C}[H]$ -submodule of $\text{Res}_H^G(V)$.

I claim that $g \cdot U$ is irreducible. Suppose that W is a $\mathbb{C}[H]$ -submodule of $g \cdot U$. Then $g^{-1} \cdot W$ is a $\mathbb{C}[H]$ -submodule of U by the above paragraph. Hence $g^{-1} \cdot W$ is U or 0 since U is irreducible. Hence W is equal to $g \cdot U$ or 0 . Thus $g \cdot U$ is irreducible. Since $g \cdot -$ is an invertible linear map, we have $\dim_{\mathbb{C}}(g \cdot U) = \dim_{\mathbb{C}}(U)$.

Let $g_1, \dots, g_l$ be complete list of distinct elements in G such that $g_1 \cdot U, \dots, g_l \cdot U$ are pairwise not equal to each other. Then $\bigoplus_{j=1}^l g_j \cdot U$ is a $\mathbb{C}[G]$ -submodule of V . Since V is an irreducible $\mathbb{C}[G]$ -module, we have $\bigoplus_{g \in G} g \cdot U = V$. Hence V is the direct sum of these irreducible submodules of equal dimension. This proves part 1 of the theorem. ■

Proposition 2.4.3

Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be an irreducible character of G . Let $\psi_1, \dots, \psi_r$ be the complete list of irreducible characters of H . Suppose that $\text{Res}_H^G \chi = \sum_{i=1}^r d_i \psi_i$. Then the following are true.

- We have $\sum_{i=1}^r d_i^2 \leq [G : H]$.
- The above equality holds if and only if $\chi(g) = 0$ for all $g \in G \setminus H$.

Proof

Notice that

$$\sum_{i=1}^r d_i^2 = \langle \text{Res}_H^G \chi, \text{Res}_H^G \chi \rangle_H = \frac{1}{|H|} \sum_{h \in H} \chi(h) \overline{\chi(h)}$$

and also

$$1 = \langle \chi, \chi \rangle_G = \frac{1}{|G|} \left(\sum_{h \in H} \chi(h) \overline{\chi(h)} + \sum_{g \in G \setminus H} \chi(g) \overline{\chi(g)} \right) = \frac{|H|}{|G|} \left(\sum_{i=1}^r d_i^2 \right) + \frac{1}{|G|} \sum_{g \in G \setminus H} \chi(g) \overline{\chi(g)}$$

so we are done. ■

Lemma 2.4.4

Let G be a group. Let k be a field. Let $H \leq G$ be a subgroup. Let $g_1H, \dots, g_nH$ be the complete list of distinct left cosets of H in G . Let U be a representation of H . Then we have

$$\text{Ind}_H^G(U) \cong \bigoplus_{i=1}^n g_i H$$

Example 2.4.5

Let $G = D_3 = \langle a, b \mid a^3, b^2, abab \rangle = S_3$. Let $H = \langle a \rangle \leq D_3$. Then the complete list of irreducible $\mathbb{C}[H]$ -modules are given by

- $W_1 = \mathbb{C}\langle 1_H + a + a^2 \rangle$.
- $W_2 = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2 \rangle$.
- $W_3 = \mathbb{C}\langle 1_H + \omega^2 a + \omega a^2 \rangle$.

where $\omega = e^{2\pi i/3}$. Then the induced representations of the modules are given by

- $\text{Ind}_H^G(W_1) = \mathbb{C}\langle 1_H + a + a^2, b + ba + ba^2 \rangle$.
- $\text{Ind}_H^G(W_2) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega ba + \omega^2 ba^2 \rangle$.
- $\text{Ind}_H^G(W_3) = \mathbb{C}\langle 1_H + \omega a + \omega^2 a^2, b + \omega^2 ba + \omega ba^2 \rangle$.

Recall that we have a change of rings adjunction formula:

$$\text{Hom}_{k[G]}(\text{Ind}_H^G(U), V) \cong \text{Hom}_{k[H]}(U, \text{Res}_H^G(V))$$

Theorem 2.4.6 (Frobenius Reciprocity Theorem)

Let G be a finite group. Let $H \leq G$ be a subgroup. Let χ be a character of G . Let ψ be a character of H . Then the change of base rings adjunction induces the following:

$$\langle \text{Ind}_H^G \psi, \chi \rangle_G = \langle \psi, \text{Res}_H^G \chi \rangle_H$$

Proposition 2.4.7 (The Mackey Formula)

Let G be a finite group. Let $H \leq G$ be a subgroup. Let ψ be a character of H . Then we have

$$\text{Ind}_H^G \psi(g) = \frac{1}{|H|} \sum_{y \in G, ygy^{-1} \in H} \psi(ygy^{-1})$$

3 Representation Theory of the Symmetric Group

3.1

The conjugacy classes of S_n is the given by the cycle type of the permutations. These are indexed by partitions $\lambda = (\lambda_1, \dots, \lambda_k)$ such that $\sum_{i=1}^k \lambda_i = n$ and $\lambda_i \geq \lambda_{i+1}$. We write $\lambda \vdash n$. These can be drawn into Young diagrams:

- There are λ_i boxes in row i .
- Rows are left justified.

Let S_λ be the subgroup of S_n given by

$$S_\lambda = S_{1, \dots, \lambda_1} \times S_{\lambda_1+1, \dots, \lambda_1+\lambda_2} \times \dots \times S_{n-\lambda_k+1, \dots, n} \cong S_{\lambda_1} \times \dots \times S_{\lambda_k}$$

Let M^λ be the module associated to the character $\text{Triv}_{S_\lambda}^{S_n}$. This is not generally irreducible. But we can order partitions of n such that the following are true.

- $M^{\lambda(1)}$ is irreducible (call it $S^{\lambda(1)}$)
- $M^{\lambda(2)}$ contain copies of $S^{\lambda(1)}$ and a unique copy of a new irreducible module $S^{\lambda(2)}$.
- etc...

$S^{\lambda(i)}$ are called Specht modules. $M^{\lambda(i)}$ are called the permutation modules corresponding to λ . The ordering needed is reverse-lexicographic (revlex). Largest number to smallest number, starting with the first component and moving to latter components to break ties.

Let $\bar{t}_i$ denote the distinct λ -tabloids: fill the young diagram of λ with $\{1, \dots, n\}$ (this is called a Young tableau) and mark two as equivalent if each corresponding row has the same elements. Then $M^\lambda = \mathbb{C}\langle \bar{t}_1, \dots, \bar{t}_k \rangle$.

Example 3.1.1

The partitions of $\{1, \dots, 4\}$ is given by

$$\{(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1)\}$$

Example 3.1.2

Let $\lambda = (n)$. Then $M^\lambda \cong \mathbb{C}$ and S_n acts on this trivially. Thus $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \text{Triv}$.

Let $\lambda = (1, \dots, 1)$. Then $M^\lambda \cong \mathbb{C}^{n!}$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv} = \chi_{\text{reg}}$.

Let $\lambda = (n-1, 1)$. Then $M^\lambda \cong \mathbb{C}^n$. Then $\text{Ind}_{S_\lambda}^{S_n} \text{Triv}$ is the character of the permutation representation.

Definition 3.1.3 (Young Diagram)

Let $n \in \mathbb{N}^+$. Let $\lambda = (\lambda_1, \dots, \lambda_k)$ be a partition of n . Define the Young diagram of λ to be

$$Y(\lambda) = \{(i, j) \in \mathbb{N}^+ \times \mathbb{N}^+ \mid 1 \leq i \leq k, 1 \leq j \leq \lambda_i\}$$

Definition 3.1.4 (Young Tableau)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n . A Young tableau of λ is a bijection $t : Y(\lambda) \rightarrow \{1, \dots, n\}$. Denote the set of all Young tableau of λ by Δ^λ .

Definition 3.1.5 (Young Tabloid)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n .

- Let t_1, t_2 be Young tableau of λ . We say that t_1 and t_2 are equivalent if

$$\{t_1(r_0, j) \mid 1 \leq j \leq \lambda_i\} = \{t_2(r_0, j) \mid 1 \leq j \leq \lambda_k\}$$

for all $1 \leq r_0 \leq k$ (in other words the rows of the tableau are equal up to permutation).

- A Young tabloid of λ is an equivalence class $[t]$ where t is a Young tableau of λ .

Definition 3.1.6 (The Associated Polytabloid of a Tableau)

Let $n \in \mathbb{N}^+$. Let λ be a partition of n . Let t be a λ -tableau. Define the associated polytabloid to be

$$e(t) = \sum_{\sigma \in C(t)} \text{sign}(\sigma)(\sigma \cdot [t])$$

where $C(t) = \prod_{i=1}^k \text{Sym}(C_i)$ where C_i is the i th column of t .

Definition 3.1.7 (The Specht Module of a Partition)

Let $n \in \mathbb{N}^+$. Let λ be a partition for n . Define the Specht module associated to λ to be the $\mathbb{C}$ -vector space

$$S^\lambda = \mathbb{C}\langle e(t) \mid t \text{ is a } \lambda\text{-tableau} \rangle$$

Theorem 3.1.8

Let $n \in \mathbb{N}^+$. Then $\{S^\lambda \mid \lambda \text{ is a partition for } n\}$ forms the complete list of non-isomorphic irreducible $\mathbb{C}[S_n]$ -modules.

4 Basic Definitions of Representations

4.1 Representation of an Algebra

Definition 4.1.1 (Representation of an Algebra)

Let R be a commutative ring. Let A be an R -algebra. A representation of A is a left R -module M and a homomorphism of R -algebras

$$\rho : A \rightarrow \text{End}_R(M)$$

Let A, M be sets. Recall that there is a natural bijection

$$\text{Hom}_{\text{Sets}}(A \times M, M) \cong \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

given by curling. A representation of an R -algebra A is an element in

$$\text{Hom}_{\text{Alg}_R}(A, \text{End}_R(M)) \subseteq \text{Hom}_{\text{Sets}}(A, \text{Hom}_{\text{Sets}}(M, M))$$

Therefore we may ask what kinds of maps of the form $A \times M \rightarrow M$ give a representation of A .

Proposition 4.1.2

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Let $f : A \times M \rightarrow M$ be a function. Then the map $a \mapsto f(a, -)$ defines a representation of A if and only if the following are true.

- f is an R -module homomorphism.
- f defines an A -module structure on M .

The A -module structure can also be seen in the representation $\rho : A \rightarrow \text{End}_R(M)$. It is given by $a \cdot m = \rho(a)(m)$. In particular, not every A -module M gives a representation; the action of A on M must be bilinear in the scalars R . Therefore these modules are often called R -linear A -modules.

However, if $R = k$ is a field and the structure map $k \rightarrow A$ is non-zero, then $k \rightarrow A$ is necessarily injective so that A contains k as a subring. Then the k -linearity of $f : A \times M \rightarrow M$ is null so that A -modules are the same as representations of A over k .

Corollary 4.1.3

Let k be a field. Let A be a k -algebra. Let V be a k -vector space. Let $f : A \times V \rightarrow V$ be a function. Then the map $a \mapsto f(a, -)$ is a representation of A if and only if f defines an A -module structure on V .

Since representations of an algebra A are A -modules, it follows that the following notions in Rings and Modules apply to representations.

- Subrepresentations.
- Quotient Representations.
- Sum of Representations.

Definition 4.1.4 (Types of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be a representation of A .

- We say that M is irreducible if it is simple as an A -module.
- We say that M is semisimple if it is semisimple as an A -module.
- We say that M is indecomposable if it is indecomposable as an A -module.

Example 4.1.5

Let k be a field. Define a representation $\rho : k[x] \rightarrow \text{End}_k(k^2) \cong M_2(k)$ by $x \mapsto \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ then extended it R -linearly and respecting products of x . Then ρ is indecomposable but reducible.

Proof

Suppose for a contradiction that ρ is decomposable. Then there exists 1-dimensional linear subspaces L_1, L_2 of k^2 such that ρ_{L_1} and ρ_{L_2} are representations. Suppose that $L_1 = k\langle v_1 \rangle$ and $L_2 = k\langle v_2 \rangle$. Then $x \cdot v_1 \in L_1$ and $x \cdot v_2 \in L_2$ implies that v_1 and v_2 are linearly independent eigenvectors of $\rho(x)$. But by an easy computation, we see that the only eigenvalue of $\rho(x)$ is 1 and its eigenspace has dimension 1. Hence this is a contradiction.

From the above, we see that the eigenspace of $\rho(x)$ is a subrepresentation of ρ . Hence ρ is reducible. ■

Definition 4.1.6 (The Regular Representation)

Let k be a field. Let A be a k -algebra. Define the regular representation $\text{reg} : A \rightarrow \text{End}_k(A)$ of A by

$$\text{reg}(a)(b) = ab$$

Definition 4.1.7 (Composition Factors)

Let R be a commutative ring. Let A be an R -algebra. Let M be a semisimple representation of A . Suppose that the irreducible decomposition of M is given by

$$M \cong \bigoplus_{i=1}^n M_i$$

Define the composition factors of M to be

$$\text{Comp}(M) = \{M_1, \dots, M_n\}$$

Definition 4.1.8 (Multiplicity of an Irreducible Submodule)

Let R be a commutative ring. Let A be an R -algebra. Let M be a semisimple representation of A . Let N be an irreducible A -module. Define the multiplicity of N in M to be

$$\text{mult}_N(M) = |\{M_i \in \text{Comp}(M) \mid M_i \cong N\}|$$

4.2 Maps between Representations

Definition 4.2.1 (Homomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be representations of A . Let $f : M \rightarrow N$ be a function. We say that f is a homomorphism of representations if the following are true.

- f is an A -module homomorphism.
- f is R -linear.

Since M and N are A -modules, the definition of homomorphism implies that such a morphism must be an A -module homomorphism. Conversely, since not every A -module is an R -linear A -module, A -module homomorphisms are not necessarily morphisms between representations.

Definition 4.2.2 (Set of Homomorphisms of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be representations of A . Define the set of homomorphisms from M to N to be

$$\text{Hom}_{A_R}(M, N) = \left\{ f : M \rightarrow N \mid \begin{array}{c} f \text{ is a homomorphism} \\ \text{of representations from } M \text{ to } N \end{array} \right\} \subseteq \text{Hom}_A(M, N)$$

Definition 4.2.3 (Isomorphisms of Representations) Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. We say that ϕ is an isomorphism if there exists a homomorphism of representations $\psi : N \rightarrow M$ such that

$$\psi \circ \phi = \text{id}_M \quad \text{and} \quad \phi \circ \psi = \text{id}_N$$

Lemma 4.2.4

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then ϕ is an isomorphism if and only if ϕ is an isomorphism of R -modules.

Theorem 4.2.5 (First Isomorphism Theorem for Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M, N be R -modules. Let $\rho : A \rightarrow \text{End}_R(M)$ and $\tau : A \rightarrow \text{End}_R(N)$ be representations of A . Let $\phi : M \rightarrow N$ be a homomorphism of representations. Then the following are true.

- $\ker(\phi)$ is a subrepresentation of M .
- $\text{im}(\phi)$ is a subrepresentation of N .
- There is an isomorphism of representations

$$M / \ker(\phi) \cong \text{im}(\phi)$$

given by ϕ .

Proposition 4.2.6

Let R be a commutative ring. Let A be a finite R -algebra. Let $\rho : A \rightarrow \text{End}_R(M)$ be an irreducible representation of A . Then M is finitely generated.

Proof

Let $0 \neq m \in M$. Then $\rho(A)(m)$ is a submodule of M . Since M is irreducible, we must have $\rho(A)(m) = M$. Since A is finite over R , we must have $\rho(A)(m)$ is finitely generated over R . Hence M is finitely generated over R . ■

4.3 The Set of Representations

Definition 4.3.1 (The Set of Representations)

Let R be a commutative ring. Let A be an R -algebra. Let M be an R -module. Define the space of representations of A on M to be the set

$$\text{Rep}(A, M) = \{[\rho : A \rightarrow \text{End}_R(M)] \mid [\rho] \text{ is an isomorphism class of representations} \}$$

5 Finite Dimensional Representations of Algebra over Algebraically Closed Fields

5.1 Schur's Lemma and its Consequences

Recall from Rings and Modules that we have Schur's lemma: A homomorphism $M \rightarrow N$ of simple R -modules is either the 0 map or an isomorphism.

Lemma 5.1.1 (Schur's Lemma III)

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Let $\phi : V \rightarrow V$ be a homomorphism of representations. Then $\phi = \lambda \text{id}_V$ for some $\lambda \in k$.

Proof

Since ϕ is a homomorphism of simple A -modules, by Schur's lemma we know that $\phi = 0$ or ϕ is an isomorphism. If $\phi = 0$ then we are done. Otherwise, since k is algebraically closed, the endomorphism ϕ admits an eigenvalue $\lambda \in k$ and eigenvector $v \in V$. Consider the homomorphism of representations $\phi - \lambda \text{id}_V$. By Schur's lemma, this map is either 0 or an isomorphism. But it is not an isomorphism since $(\phi - \lambda \text{id}_V)(v) = 0$. Hence $\phi - \lambda \text{id}_V = 0$. So we are done. ■

Corollary 5.1.2

Let k be an algebraically closed field. Let A be a commutative k -algebra. Let V be a finite dimensional k -vector space. Let $\rho : A \rightarrow \text{End}_k(V)$ be an irreducible representations of A . Then $\dim(V) = 1$.

Proof

By Schur's lemma III, we know that $\rho(a) : V \rightarrow V$ is a scalar operator for all $a \in A$. This means that $\rho(a)(v) = \lambda_a v$ for some $\lambda_a \in k$ and all $v \in V$. For any fixed $v_0 \in V$, this implies that $k\langle v_0 \rangle$ is a subrepresentation of ρ . If $\dim(V) \geq 2$, then $k\langle v_0 \rangle$ is a non-trivial subrepresentation, contradicting the irreducible assumption. Hence $\dim(V) = 1$. ■

In other words, every finite dimensional representation of a commutative algebra over an algebraically closed field must be of dimension 1. And these are easy to classify.

Proposition 5.1.3

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be finite dimensional irreducible representations of A . Then we have

$$\dim_k(\text{Hom}_{A_k}(V, W)) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Proof

If V is not isomorphic to W , then by Schur's lemma we must have $\text{Hom}_{\text{Rep}(A)}(V, W) \subseteq \text{Hom}_A(V, W) = \{0\}$. If $V = W$, then Schur's lemma III implies that id_V generates the k -vector space. ■

Proposition 5.1.4

Let k be an algebraically closed field. Let A be a k -algebra. Let V_i, W_j be finite dimensional irreducible representations of A for $1 \leq i \leq n$ and $1 \leq j \leq m$. Then we have

$$\dim_k \left(\text{Hom}_{A_k} \left(\bigoplus_{i=1}^n V_i, \bigoplus_{j=1}^m W_j \right) \right) = \sum_{i=1}^n \sum_{j=1}^m \dim_k(\text{Hom}_{A_k}(V_i, W_j))$$

Corollary 5.1.5

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be finite dimensional representations of A such that W is irreducible. Then we have

$$\dim_k(\operatorname{Hom}_{A_k}(V, W)) = \operatorname{mult}_W(V)$$

5.2 Character Theory

Let k be an algebraically closed field. Recall that the trace of a k -linear map between finite dimensional vector spaces have a basis independent description.

Definition 5.2.1 (Characters of a Representation)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \operatorname{End}_k(V)$ be a finite dimensional representation of A . Define the character of ρ to be function $\chi_V : A \rightarrow k$ given by

$$\chi_V(a) = \operatorname{tr}(\rho(a))$$

Proposition 5.2.2

Let k be an algebraically closed field. Let A be a k -algebra. Let V, W be isomorphic finite dimensional representations of A . Then $\chi_V = \chi_W$.

Proposition 5.2.3

Let k be an algebraically closed field. Let A be a k -algebra. Let $\rho : A \rightarrow \operatorname{End}_k(V)$ be a finite dimensional representation of A . Then χ_V descends to a well defined function $\chi_V : \frac{A}{[A, A]} \rightarrow k$.

Let G be a finite group. Since $\mathbb{C}[G]$ is a semisimple $\mathbb{C}$ -algebra, we know that there are $\mathbb{C}$ -vector space isomorphisms

$$\begin{aligned} \operatorname{Hom}_{\mathbb{C}}\left(\frac{\mathbb{C}[G]}{[\mathbb{C}[G], \mathbb{C}[G]]}, \mathbb{C}\right) &= \{f \in \operatorname{Hom}_{\mathbb{C}}(\mathbb{C}[G], \mathbb{C}) \mid f(ghg^{-1}h^{-1}) = 0 \text{ for all } g, h \in G\} \\ &= \{f \in \operatorname{Hom}_{\operatorname{Set}}(G, \mathbb{C}) \mid f(ghg^{-1}) = f(h) \text{ for all } g, h \in G\} \\ &= \operatorname{Hom}_{\operatorname{Set}}(\operatorname{Cl}(G), \mathbb{C}) \end{aligned}$$

Hence the above proposition mimics character theory in representations of finite groups.

Lemma 5.2.4

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a finite dimensional representation of A . Let $W \leq V$ be a subrepresentation. Then we have

$$\chi_V = \chi_W + \chi_{V/W}$$

Definition 5.2.5 (Irreducible Characters)

Let k be an algebraically closed field. Let A be a k -algebra. Let V be a representation of A . We say that χ_V is irreducible if V is irreducible.

Proposition 5.2.6

Let k be an algebraically closed field. Let A be a k -algebra. Let $V_1, \dots, V_n$ be finite dimensional irreducible representations of A . Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ are linearly independent in $\operatorname{Hom}_k(A/[A, A], k)$.

Definition 5.2.7 (Characters of an Algebra)

Let k be an algebraically closed field. Let A be a k -algebra. Let $\phi : A \rightarrow k$ be a function. We say that ϕ is a character of A if there exists a finite dimensional representation V of A such that $\phi = \chi_V$.

5.3 The Semisimple Condition

Proposition 5.3.1

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A . Then there is a k -algebra isomorphism

$$A \cong \bigoplus_{i=1}^n \text{End}_k(V_i)$$

Proof

By the Artin Wedderburn theorem, there is a k -algebra isomorphism

$$A \cong \bigoplus_{i=1}^p M_{n_i}(D_i)$$

where each D_i is a division algebra over k . Since k is algebraically closed, we have $D_i = k$. By a corollary of the Artin-Wedderburn theorem, the complete list of simple A -modules is given by the unique simple $M_{n_i}(k)$ -submodule T_i of itself. Hence $n = p$. Moreover, by reordering the index, we have isomorphisms of $M_{n_i}(k)$ -modules $V_i \cong T_i$ that are isomorphisms of A -modules (since the ring structure of the product of matrices is given by the product of the ring structures of the matrix ring, and the Artin-Wedderburn isomorphism is a ring isomorphism). Now $T_i \cong k^{n_i}$ as k -vector spaces. This gives an isomorphism $V_i \cong k^{n_i}$ of k -vector spaces and hence an isomorphism of rings $\text{End}_k(V_i) \cong M_{n_i}(k)$. Thus we have

$$A \cong \bigoplus_{i=1}^p M_{n_i}(k) \cong \bigoplus_{i=1}^n \text{End}_k(V_i)$$

■

Corollary 5.3.2

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic representations of A . Then we have

$$\dim_k(A) = \sum_{i=1}^n \dim_k(V_i)^2$$

Corollary 5.3.3

Let k be an algebraically closed field. Let A be a finite k -algebra. Let V be an irreducible representation of A . Then we have

$$\text{mult}_V(A) = \dim_k(\text{Hom}_{A_k}(A, V)) = \dim_k(V)$$

Proof

Direct from Artin-Wedderburn theorem and its corollaries. ■

Proposition 5.3.4

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let $V_1, \dots, V_n$ be the complete list of non-isomorphic irreducible finite dimensional representation of A . Then $\{\chi_{V_1}, \dots, \chi_{V_n}\}$ is a basis for $\text{Hom}_k(A/[A, A], k)$.

Corollary 5.3.5

Let k be an algebraically closed field. Let A be a finite semisimple k -algebra. Let χ be a character of A . Then there exists a representation V of A such that $\chi = \chi_V$.

5.4 Reduction to Indecomposable Basic Algebras

Let A be a finite dimensional k -algebra over a field k . Recall that $A^b = eAe$ is the basic algebra associated to A , where $e = \sum_{i=1}^n e_i$ and $e_1, \dots, e_n$ is a full orthogonal system of primitive idempotents.

Proposition 5.4.1

Let k be an algebraically closed field. Let A be a finite dimensional k -algebra. Then there is an equivalence of categories

$${}_A \mathbf{FGMod} \simeq {}_{A^b} \mathbf{FGMod}$$

given by the restriction and extension of scalars.

Proposition 5.4.2

Let k be an algebraically closed field. Let A, B be k -algebras. Let V be a finite dimensional irreducible representation of $A \times B$. Then there exists irreducible representations V_A and V_B of A and B respectively, such that $V \cong V_A \otimes_k V_B$.

6 Quivers and Path Algebras

6.1 Basics of Quivers

Definition 6.1.1 (Quivers)

A quiver consists of two sets V and E called vertices and edges, together with two functions $s, t : E \rightarrow V$ called the source and target respectively.

Definition 6.1.2 (Finite Quivers)

Let $Q = (V, E, s, t)$ be a quiver. We say that Q is finite if V and E are finite.

Definition 6.1.3 (Connected Quivers)

Definition 6.1.4 (Paths and Stationary Path)

Let $Q = (V, E, s, t)$ be a quiver. Let $a, b \in V$. A path from a to b of length $n \in \mathbb{N}^+$ is a function $\gamma : \{1, \dots, n\} \rightarrow E$ such that the following are true.

- $s(\gamma(1)) = a$.
- $s(\gamma(i+1)) = t(\gamma(i))$ for $1 \leq i \leq n-1$.
- $t(\gamma(n)) = b$.

For each $v \in V$, we associate a path ε_v of length 0 called the stationary path at v .

Definition 6.1.5 (Cycles)

Let $Q = (V, E, s, t)$ be a quiver. Let $\gamma : \{1, \dots, n\} \rightarrow E$ be a path in Q . We say that γ is a cycle if $s(\gamma(1)) = t(\gamma(n))$.

Definition 6.1.6 (Acyclic Quivers)

Let Q be quiver. We say that Q is acyclic if Q contains no cycles.

6.2 Path Algebra

Definition 6.2.1 (The Path Algebra of a Quiver)

Let Q be a quiver. Let k be a field. Define the path algebra $k[Q]$ of Q over k to be the k -vector space

$$k\langle Q \rangle = k\langle \gamma \mid \gamma \text{ is a path in } Q \rangle$$

together with multiplication given as follows. For two paths $\gamma_1 : \{1, \dots, n_1\} \rightarrow E$ and $\gamma_2 : \{1, \dots, n_2\} \rightarrow E$, define $\gamma_1 \cdot \gamma_2 : \{1, \dots, n_1 + n_2\} \rightarrow E$ by

$$(\gamma_1 \cdot \gamma_2)(i) = \begin{cases} \gamma_1(i) & \text{if } 1 \leq i \leq n_1 \\ \gamma_2(i - n_1) & \text{if } n_1 + 1 \leq i \leq n_2 \end{cases}$$

if $t(\gamma_1(n_1)) = s(\gamma_2(1))$ and 0 otherwise.

Lemma 6.2.2

Let $Q = (V, E, s, t)$ be a quiver. Let k be a field. If V is finite, then the following are true.

- $k\langle Q \rangle$ is a graded k -algebra whose grading is given by the length of the paths.
- We have $1_{k\langle Q \rangle} = \sum_{v \in V} \varepsilon_v$.
- $\{\gamma : \{1\} \rightarrow E \mid \gamma \text{ is a path in } Q\}$ is a generating set for $k\langle Q \rangle$ as a module over itself.
- $\{\varepsilon_v \mid v \in V\}$ forms a full orthogonal system of primitive idempotents for $k\langle Q \rangle$.

Example 6.2.3

Let $Q = (V, E, s, t)$ be the quiver given by $V = \{v\}$ and $E = \{\alpha_i \mid i \in I\}$. Then we have a k -algebra

isomorphism

$$k\langle Q \rangle \cong k\langle \alpha_i \mid i \in I \rangle$$

to the free associative algebra generated by E .

Proposition 6.2.4

Let Q be a quiver. Let k be a field. Then the following are true.

- $k\langle Q \rangle$ is finite dimensional over k if and only if Q is finite and acyclic.
- $k\langle Q \rangle$ is indecomposable if and only if Q is connected.

Proposition 6.2.5 (Universal Property of Finite Connected Quivers Algebras)

Let $Q = (V, E, s, t)$ be a finite connected quiver. Let k be a field. Then $k\langle Q \rangle$ and the inclusion functions $\iota_V : V \rightarrow k\langle Q \rangle$ and $\iota_E : E \rightarrow k\langle Q \rangle$ satisfies the following universal property: For any k -algebra A and any maps $\varphi_V : V \rightarrow A$ and $\varphi_E : E \rightarrow A$ such that the following are true:

- $1 = \sum_{v \in V} \varphi_V(v)$, $\varphi_V(v)^2 = \varphi_V(v)$ and $\varphi_V(v)\varphi_V(w) = 0$ for all $v, w \in V$ and $v \neq w$.
- We have $\varphi_E(p) = \varphi_V(s(p))\varphi_E(p)\varphi_V(t(p))$ for all $p \in E$.

there exists a unique k -algebra homomorphism $\varphi : k\langle Q \rangle \rightarrow A$ such that the following diagram commutes:

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & k\langle Q \rangle & \xleftarrow{\iota_E} & E \\ & \searrow \varphi_V & \downarrow \exists! \varphi & \swarrow \varphi_E & \\ & & A & & \end{array}$$

Moreover, $k\langle Q \rangle$ is the unique (up to unique isomorphism) k -algebra with the above property.

Definition 6.2.6 (The Arrow Ideal)

Let Q be a finite quiver. Let k be a field. Define the arrow ideal of the path algebra $k\langle Q \rangle$ to be

$$R_Q = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length at least } 1 \rangle$$

Lemma 6.2.7

Let Q be a finite quiver. Let k be a field. Then the following are true.

- R_Q is a two sided graded ideal of $k\langle Q \rangle$.
- The quotient of the path algebra by the arrow ideal is given by

$$k\langle Q \rangle / R_Q = k\langle \varepsilon_v \mid v \in V \rangle$$

- For each $n \in \mathbb{N}^+$, the product of the arrow ideal is given by

$$R_Q^n = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length at least } n \rangle$$

- For each $n \in \mathbb{N}^+$, we have

$$R_Q^n / R_Q^{n+1} = k\langle \gamma \mid \gamma \text{ is a path in } Q \text{ of length } n \rangle$$

Proposition 6.2.8

Let Q be a finite quiver. Let k be a field. If Q is acyclic, then the following are true.

- $J(k\langle Q \rangle) = R_Q$.
- $k\langle Q \rangle$ is a finite dimensional basic algebra over k .

Definition 6.2.9 (Admissible Ideals)

Let Q be a finite quiver. Let k be a field. Let I be an ideal of $k\langle Q \rangle$. We say that I is admissible if there exists $n \geq 2$ such that

$$R_Q^n \subseteq I \subseteq R_Q^2$$

Definition 6.2.10 (Bound Quiver Algebra)

Let Q be a finite quiver. Let k be a field. Let I be an ideal of $k\langle Q \rangle$. We say that $k\langle Q \rangle/I$ is a bound quiver algebra if I is an admissible ideal.

Proposition 6.2.11

Let $Q = (V, E, s, t)$ be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q \rangle$. Then $\{\varepsilon_v + I \mid v \in V\}$ is a full orthogonal system of primitive idempotents for the bound quiver algebra $k\langle Q \rangle/I$.

Proposition 6.2.12

Let Q be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q \rangle$. Then the following are true.

- I is finitely generated.
- $J(k\langle Q \rangle/I) = R_Q/I$.
- $k\langle Q \rangle/I$ is a finite dimensional basic algebra over k .
- $k\langle Q \rangle/I$ is indecomposable if and only if $k\langle Q \rangle$ is indecomposable.

6.3 The Quiver of a Finite Dimensional Basic Algebra

Definition 6.3.1 (The Quiver of a Basic Algebra)

Let k be a field. Let A be a finite dimensional basic algebra over k . Let $e_1, \dots, e_n$ be a full orthogonal system of primitive idempotents. Define the quiver $Q_A = (V, E, s, t)$ associated to A as follows.

- The vertices are given by $V = \{e_1, \dots, e_n\}$.
- For $e_i, e_j \in V$, the set of arrows from e_i to e_j are given by $e_i \frac{J(A)}{J^2(A)} e_j$.

Lemma 6.3.2

Let k be a field. Let A be a finite dimensional basic algebra over k . Then Q_A does not depend on the choice of full orthogonal system of primitive idempotents.

Theorem 6.3.3 (Gabriel)

Let k be an algebraically closed field. Let A be a finite dimensional basic algebra over k . Then there exists an admissible ideal I of $k\langle Q_A \rangle$ such that there is a k -algebra isomorphism

$$A \cong \frac{k\langle Q_A \rangle}{I}$$

Corollary 6.3.4

Let k be an algebraically closed field. Let A be a finite dimensional algebra over k . Then A is Morita equivalent to $k\langle Q_{A^\flat} \rangle/I$ where I is an admissible ideal of $k\langle Q_{A^\flat} \rangle$.

Thus the representation theory of a finite dimensional algebra over an algebraically closed field is equivalent to the representation theory of bound finite quiver algebras.

7 Representation Theory of Quivers

7.1 Representations of Quivers

Definition 7.1.1 (The Category of Representations of a Quiver)

Definition 7.1.2 (The Category of Representations of a Quiver over an Admissible Ideal)

Let k be a field. Let Q be a quiver. Let I be an admissible ideal of the path algebra $k\langle Q \rangle$. Define $\mathbf{Rep}_I(Q)$ to be the full subcategory of $\mathbf{Rep}(Q)$ consisting of objects $((V_i)_{i \in V}, (T_j)_{j \in E})$ such that $V_k = 0$ for all $k \in I$.

7.2 Relation to Representations of the Path Algebra

Proposition 7.2.1

Let Q be a finite quiver. Let k be a field. Let I be an admissible ideal of $k\langle Q \rangle$. Then the following are true.

- There